



Water powered spacecraft!

A tiny, water powered space craft could be the first to mine asteroids.

<http://dgit.in/waterpoweredspacecraft>



Cloud-seeding drones!

"Sandoval Silver State Seeder" is a new drone which de-ploys silver iodide flares to kick off rainfall. http://dgit.in/cloud_seeding

A full scale model of the James Webb Space Telescope on display in Austin, Texas in 2013



HOW WILL THE JAMES WEBB SPACE TELESCOPE HELP US PEER BETTER INTO THE DARKNESS OF SPACE?

The Hubble's successor sure is worth looking at.

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Even if you don't work in astronomy, cosmology or any other allied branch of physics or engineering, odds are that you might have heard of the Hubble Space Telescope (HST), such has been the magnitude of its success. HST has been NASA's flagship mission for a very long time now, gathering critical data from regions in the cosmos light years away from the earth and helping scientists unravel the many mysteries of our universe. But, believe it or not, even though the Hubble is a landmark feat of advanced engineering, it too can become obsolete. HST has been on mission for more than two decades now, and while it still has functioning capabilities

many scientists believe that the HST is now incapable of pushing the limit of astronomical observations. Astronomers love data. Colorful images are great for public relations but scientists need hard data to study the workings of the Universe. They want to capture light from the early universe, study structure formations that currently can't be studied and in general push the envelope for what we can see and what is in the dark, as of now. Enter, the James Webb Space Telescope (JWST), Hubble's worthy successor, NASA's next flagship mission, a ~9 billion \$ beast of a device and a telescope of unprecedented versatility and sensitivity.

JWST: HST 2.0?

There is no denying that the HST ushered us into a new age of science. It gave theorists data points to fit their theory and allowed experimentalists to make observations without worrying about the many obstacles that hampered earth

based telescopes and limited their range. While scientists still routinely employ the HST (it is usually overbooked) for their studies, the JWST opens up new avenues for more detailed measurements. What gives the JWST the edge? For starters, the JWST has a much larger mirror than the HST. The primary mirror of a telescope serves as a light collecting device, making it an extremely critical component. The larger primary mirror of the HST allows it to have greater area for collection (7 x HST) and effectively a larger field of view (~15 x HST) for the camera at the heart of the device. This in turn enables the JWST to have extremely high spatial resolution. While we are on the subject of the primary mirrors, the one installed in the JWST is arguably the largest to be ever launched into space.

A significant amount of innovation has gone into fabricating mirrors of such dimensions and sophistication. Another complication that had to be accounted for is that the mirrors had to be kept cold, really cold, at a cryogenic state of around -220 Degree Celsius! The mirrors in the HST, while coated with reflective materials, still face heat from the sun continuously, increasing their temperature. This imposes a limitation on the range of wavelengths the telescope can 'see'. The higher the temperature of the mirror the more vigorous the random jiggling motion of its atoms. If light coming from an observed galaxy happens to be at the lower end of the energy spectrum (high wavelength), this light signal might get lost amidst all that 'noise'. This is the limitation imposed by the design of the HST, and the JWST is made to overcome that very limitation. Not only is the JWST proposed



Pictorial comparison of the primary mirrors in the HST and JWST